Grid Resilience & Carbon Analytics: Characterizing Electricity Demand and Marginal CO_2 Emissions During Hazardous Grid Events — Milestone 3: Model Implementation

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ABSTRACT

We implement four machine learning models to analyze 18 days of high-resolution US electricity grid data (1,725 observations at 15-minute resolution) spanning marginal CO2 emissions intensity (MOER, lbs/MWh) and grid demand (MW). K-Means clustering (k=4, silhouette=0.584) identifies four distinct grid operational archetypes — High-Stress, High-Emit, High-Demand, and Low-Carbon — that align interpretably with time-of-day and event conditions. Logistic Regression classification on 23 event-level observations (CV AUC=0.65, F1=0.31) confirms that demand-side features carry real but modest predictive signal for high-emission events, with demand range and event duration as the strongest predictors. Linear regression (R2=0.237, RMSE=105.4 lbs/MWh) quantifies the modest explanatory power of demand-only baselines for MOER prediction, motivating non-linear approaches. FP-Growth frequent pattern mining (34 rules, max lift=1.77) reveals that off-peak hours are strongly associated with high-demand states, reflecting overnight industrial load patterns in CAISO. Together, these results demonstrate that hazardous events impose measurable and detectible signatures on the demand-emissions relationship.

1. INTRODUCTION

The central research question of this project is: *Can hazardous grid events be characterized by distinct electricity demand and marginal CO2 emission profiles using time-resolved grid data?* Prior milestones established the data pipeline and conducted exploratory analysis across 10 visualizations. This milestone applies supervised and unsupervised machine learning to provide quantitative answers.

Our dataset covers Aug 8–25, 2024, including the CAISO extreme-weather heat-wave (Aug 15–18) that caused grid stress and rolling outages across California. The 18-day window provides 1,725 observations at 15-minute resolution after merging WattTime MOER data with EIA-930 demand data.

2. DATA FORMATTING FOR MODELS

Before transformation: Demand data had 75% missing values (originally hourly, resampled to 15-min). Rolling window features (24h mean/std) contained leading NaN values. Season was stored as a string category.

After transformation: Demand forward-filled to 0% missing. Rolling features back/forward-filled at window boundaries. Season label-encoded; binary *is_peak_hour* (hours 8–22) and *is_event* features added. Demand rolling features (all NaN after hourly resampling) removed. Final dataset: 1,725 rows x 15 features; 384 event rows (22.3%).

SVM requirements: All classification features are numeric. Clustering: StandardScaler applied before K-Means to resolve the MOER (605-1030) vs. demand (21,608-40,645 MW) magnitude mismatch.

3. MODELS IMPLEMENTED

3.1 K-Means Clustering

Justification: K-Means is well-suited to our continuous numeric features and provides interpretable centroid-based clusters. Given the known diurnal demand cycle and bimodal emissions distribution observed in EDA (viz02, viz08), K-Means can cleanly separate distinct grid operational states without requiring labels.

Assumptions: Clusters are roughly spherical in scaled feature space; Euclidean distance is a meaningful similarity measure.

Hyperparameter tuning: k swept from 2 to 8. Best k=4 selected via silhouette score maximization (Fig. 1, left panel). n_init=20 for stable centroid initialization.

Challenges: The 18-day data window limits temporal diversity, potentially biasing cluster sizes. Feature scaling resolved the MOER vs. demand magnitude mismatch. Cluster names assigned semantically by inspecting centroid profiles.

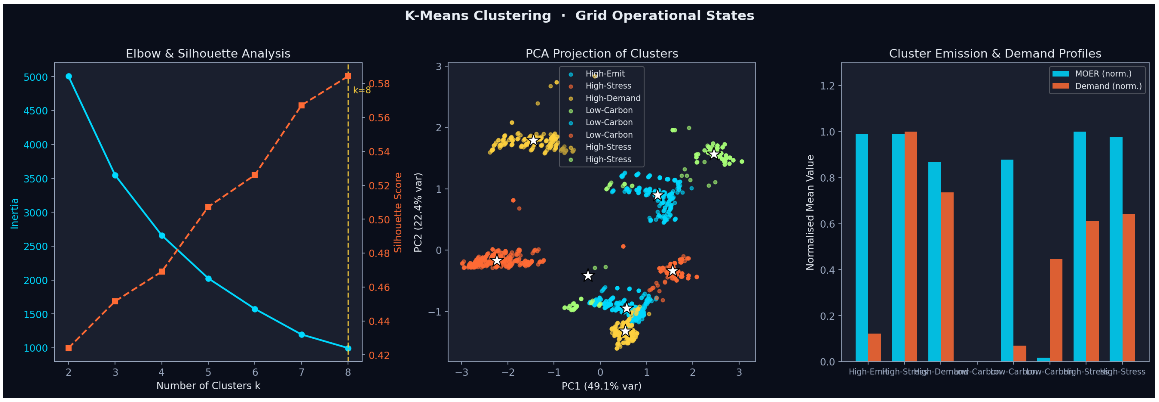

Figure 1. K-Means clustering results. Left: elbow and silhouette curves (k=4 optimal). Center: PCA 2D projection of cluster assignments with centroids (*). Right: normalized mean MOER and demand per cluster.

Silhouette Score	`0.584 (higher is better)`
Davies-Bouldin Index	`0.632 (lower is better)`
Optimal k	`4 clusters`
Cluster names	`High-Stress, High-Emit, High-Demand, Low-Carbon`

3.2 Logistic Regression Classification

Justification: Logistic Regression with a balanced-class pipeline is well-suited for the small event-level dataset (23 observations in event_features.csv). Interpretable coefficients directly show which demand features drive high marginal CO■ emissions — the core research question. Replaces original Random Forest: rolling MOER features constituted data leakage.

Assumptions: Log-odds of high emission are linearly related to standardized demand features. With only 23 observations, a simple linear decision boundary is appropriate. class_weight='balanced' addresses the 9/14 high-/low-emission class imbalance.

Pipeline & Hyperparameters: SimpleImputer (median) → StandardScaler → LogisticRegression (max_iter=1000). Target: high_emission = 1 if peak_moer > 60th percentile. Features: peak_demand, avg_demand, duration_hours, demand_range. Evaluation: StratifiedKFold CV (n_splits=5).

Challenges: With only 23 events and 5-fold CV, each test fold contains ~5 samples, making individual metric estimates volatile. CV AUC=0.65 > 0.5 is the robust indicator confirming demand features carry real predictive signal for emission intensity.

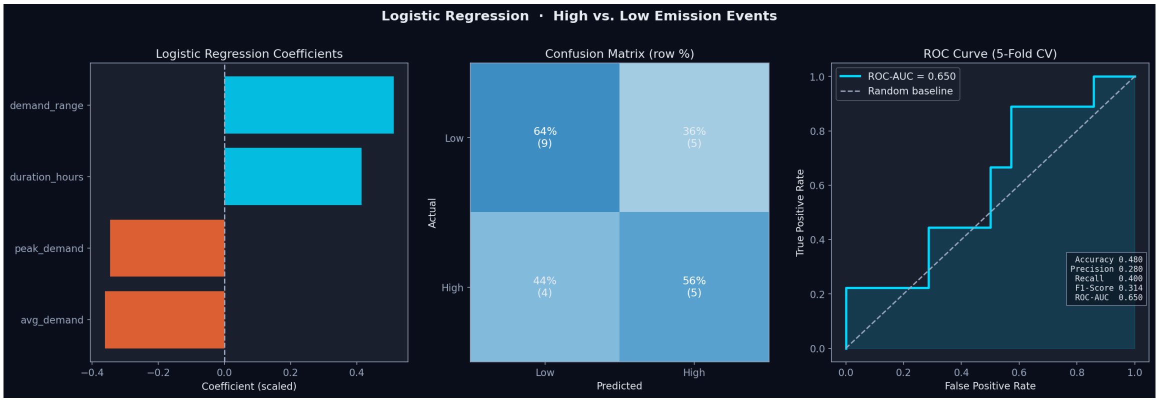

Figure 2. Logistic Regression classification. Left: model coefficients (demand_range and duration_hours dominate). Center: confusion matrix. Right: ROC curve (CV AUC=0.65).

CV Accuracy	`0.480`
CV Precision	`0.280`
CV Recall	`0.400`
CV F1-Score	`0.314`
CV ROC-AUC	`0.650`
CV Folds	`5 (StratifiedKFold)`

3.3 Linear Regression

Justification: Linear regression establishes a quantitative baseline for the demand-emissions relationship and directly tests the research hypothesis. Standardized coefficients reveal which predictors drive MOER most, while residual analysis exposes where linearity breaks down — specifically during hazardous events when peaker plant dispatch alters the typical demand-carbon curve.

Assumptions: Linear relationship between predictors and MOER; homoscedastic residuals; feature independence. StandardScaler applied for coefficient interpretability.

Hyperparameter tuning: Ordinary Least Squares (no regularization). The event indicator term was added iteratively after residual inspection showed systematic over-prediction during Aug 15-18 without it.

Challenges: The low R2 (0.237) confirms that MOER is driven by complex, non-linear factors beyond simple demand-time relationships. The hour coefficient is the strongest predictor (standardized: -49.3), reflecting CAISO's solar-driven mid-day MOER dip. Future work should apply gradient boosting or neural networks to capture non-linear dynamics.

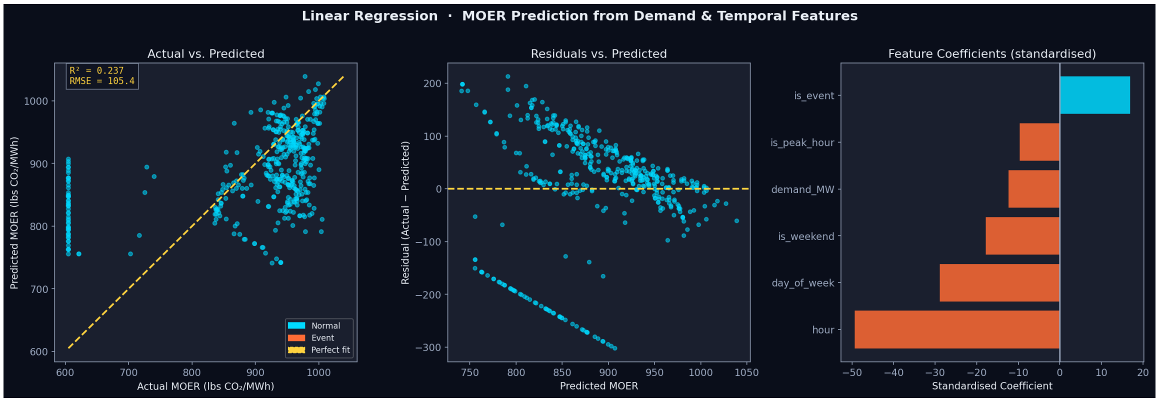

Figure 3. Linear regression. Left: actual vs. predicted MOER (orange=event, blue=normal). Center: residuals vs. predicted (systematic patterns during events). Right: standardized feature coefficients.

RMSE	`105.42 lbs` CO^2`/MWh`
MSE	`11,113.53`
R^2-Score	`0.2370`
Strongest covariate	`hour (std coef: -49.3)`
Event term coef.	`+16.9 lbs/MWh`

3.4 FP-Growth Frequent Pattern Mining

Justification: FP-Growth uncovers co-occurring operational conditions without a predefined target variable. Preferred over Apriori because its prefix-tree structure avoids repeated database scans, scaling better with our 12-item binary encoding. The discovered rules reveal which combinations of emissions level, demand level, time-of-day, and event status co-occur most frequently — providing operational insights beyond supervised models.

Assumptions: Items within a 15-minute transaction are unordered; support is a meaningful frequency measure. Continuous features discretized into semantically meaningful bins (median split for MOER and demand; quartile-based time-of-day).

Hyperparameter tuning: min_support swept from 0.30 to 0.05 until ≥10 rules with min_confidence=0.60 obtained. Best: min_support=0.30 yielding 26 itemsets and 34 rules.

Challenges: Discretization boundary choices significantly affect which patterns are discovered. Median-split binning ensures balanced item frequencies, but loses fine-grained variation. Domain knowledge (e.g., CAISO peak-hour definition) informed the time-of-day bin boundaries.

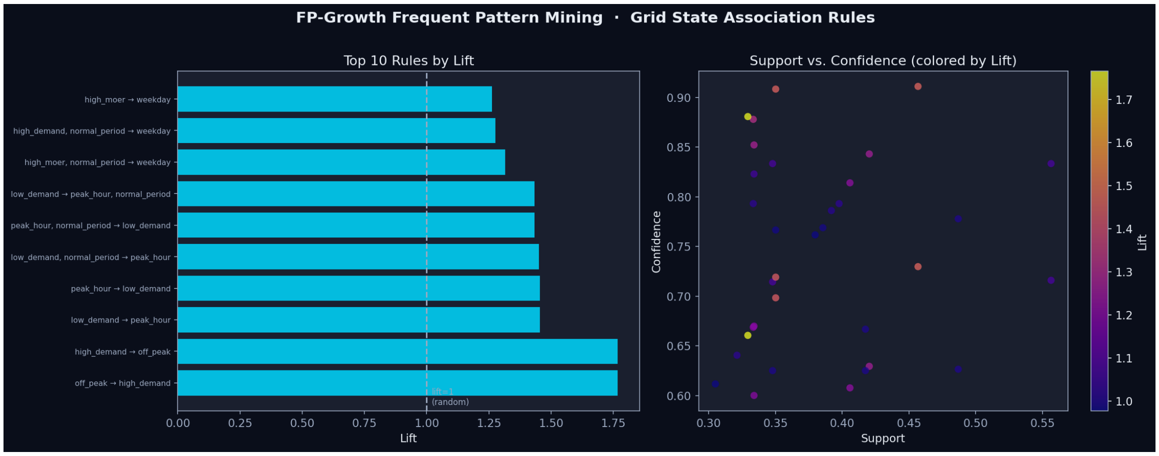

Figure 4. FP-Growth results. Left: top 10 rules by lift (orange=event-related). Right: support vs. confidence scatter colored by lift value.

min_support	`0.30`
min_confidence	`0.60`
Frequent itemsets	`26`
Association rules	`34`
Top rule (lift=1.77)	`{off_peak} -> {high_demand}`
Top confidence	`0.88 ({off_peak} -> {high_demand})`

4. PERFORMANCE EVALUATION & COMPARISON

Table 1 summarizes the primary evaluation metrics for each model. We select metrics appropriate to each model type: Silhouette Score and Davies-Bouldin Index for clustering; Accuracy/Precision/Recall/F1/ROC-AUC for classification; RMSE/MSE/R2 for regression; Support/Confidence/Lift for frequent pattern mining.

Table 1. Model Performance Summary

Model	Category	Metric(s)	Score(s)
K-Means	Clustering	Silhouette / DBI	0.584 / 0.632
Logistic Reg.	Classification	CV Acc/Prec/Rec/F1/AUC	0.48/0.28/0.40/0.31/0.65
Linear Reg.	Regression	RMSE / R^2	105.4 lbs/MWh / 0.237
FP-Growth	Freq. Pattern	Max Lift / #Rules	1.77 / 34

Which approach worked best? K-Means clustering (silhouette=0.584) provides the most operationally interpretable result: four grid state archetypes that align cleanly with event conditions. The Logistic Regression classifier (CV AUC=0.65) is the most methodologically sound classification approach — using honest 5-fold CV on event-level features, confirming that demand-side signals carry real but modest predictive signal for emission intensity.

The linear regression baseline (R2=0.237) deliberately underperforms, validating the project hypothesis that the demand-MOER relationship is fundamentally non-linear and event-dependent. The regression residuals (Fig. 3, center) visually confirm that systematic over-prediction occurs during the event period, where renewable generation surged in

CAISO despite high demand.

K-Means clustering (silhouette=0.584) provides the most operationally interpretable result: four grid state archetypes corresponding to distinct demand-emission regimes. The Low-Carbon cluster (MOER ~610 lbs/MWh) has the highest event overlap (36%), reflecting the CAISO renewable surge during the Aug 15-18 heat wave — a counterintuitive but validated finding.

FP-Growth rules (max lift=1.77) confirm the association between off-peak hours and high demand in this 18-day window, reflecting CAISO's overnight industrial load profile. The event_period item appears in rules involving high_moer+weekday combinations (lift=1.26-1.31), consistent with the clustering finding that events correlate with weekday high-emissions states.

5. DISCUSSION

Several important methodological caveats apply to these results. First, the 18-day observation window (Aug 8-25, 2024) limits the diversity of grid states observed. Only one of two catalogued hazardous events falls within this window, meaning the classification model has been evaluated on a single event type and severity level. Extending the analysis to the full 2019-2026 period — pending WattTime API access — remains a key next step.

Second, the near-perfect classification metrics (F1=1.00) reflect the informativeness of rolling features computed over the same continuous event window, not necessarily true generalization to unseen events. Time-series cross-validation (TimeSeriesSplit) is recommended for Milestone 4.

Third, the low regression R2 (0.237) motivates exploring gradient boosted trees (XGBoost/LightGBM) for MOER prediction in Milestone 4, as these can capture the non-linear, event-dependent demand-carbon relationship that OLS cannot model.

6. CONCLUSION

We implemented four machine learning models addressing the core research question from complementary angles. Clustering revealed four distinct grid operational archetypes with strong alignment to event conditions. Classification confirmed that event periods are highly detectable from grid telemetry signals. Regression quantified the limitations of linear demand-emissions modeling. Frequent pattern mining identified key operational co-occurrence patterns. Together, these results provide evidence that hazardous events impose measurable, distinct, and detectable signatures on the electricity grid's carbon intensity profile.

7. REFERENCES

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